

LearnVault AI

Professional Prompt Guide for Learners

Context engineering, prompt writing, product thinking and Lovable build prompts

Purpose of this guide

This guide helps learners understand how to write better prompts, how to give AI the right context, and how to build LearnVault AI step by step using planning prompts and Lovable build prompts.

Quick Overview

Section	What learners learn	Why it matters
Prompt basics	What a prompt is and how AI reads instructions	Learners stop writing vague prompts
Context engineering	How to give AI useful background and constraints	AI gives more accurate output
Planning prompts	Validate problem, user, root cause, features and spec	Build the right product
Lovable prompts	Build dashboard, notes, AI summary, tasks, quiz and assistant	Turn thinking into a working app
Testing prompts	Check flows, UI, database and AI output	Make the demo reliable

1. What Is a Prompt?

A prompt is the instruction you give to an AI tool. It tells the AI what you want, why you want it, what context it should use, and how the answer should be formatted.

Simple meaning

A prompt is not just a question. A good prompt is a clear instruction with context, goal, constraints and expected output.

Weak Prompt vs Strong Prompt

Weak prompt	Strong prompt
Make an app.	Build an AI second brain app for learners where users can save session notes, generate AI summaries, extract action items, search notes and ask questions from saved notes. Use a modern dashboard UI.
Summarize this.	Summarize these workshop notes for a beginner learner. Give a short summary, key points, action items and revision questions.
Give features.	List possible features for LearnVault AI and prioritize them for Version 1 using Must-have, Should-have, Could-have and Won't-have.

2. What Is Context Engineering?

Context engineering means giving AI the right background information so it can produce a better answer. It is not about writing longer prompts. It is about giving useful context in a structured way.

Important idea

Prompting tells AI what to do. Context engineering gives AI enough information to do it correctly.

The 6 Context Blocks

Block	What to include	Example for LearnVault AI
Role	Who AI should act as	Act as a product strategist and AI app builder
Goal	What you want to achieve	Create a V1 product plan for LearnVault AI
User	Who the product is for	Non-technical learners attending workshops and online courses
Problem	What pain you are solving	Learners save notes but cannot find, revise or apply them later
Constraints	Rules and limits	Keep V1 simple, buildable in Lovable, avoid complex integrations
Output format	How the answer should look	Give sections, tables, bullets and a final recommendation

3. Simple Prompt Formula

Act as [ROLE].

I am building [PRODUCT / TASK].

Context:

[Give background, user, problem and constraints]

Goal:

[Explain the outcome you want]

Please give me:

1. [Output item 1]
2. [Output item 2]
3. [Output item 3]

Rules:

- Keep it simple
- Use practical examples
- Do not add unnecessary features
- Format the answer clearly

Example

Act as a product strategist and AI app builder.

I am building LearnVault AI, an AI second brain for learners.

Context:

Learners attend courses, workshops and online sessions. They save notes, links, screenshots and prompts, but later cannot find, revise or apply what they learned.

Goal:

Help me define the first version of this product.

Please give me:

1. Target user
2. Main problem
3. Core product promise
4. Version 1 features
5. Simple user flow

Rules:

Keep it simple, practical and beginner-friendly.

4. Planning Prompts for LearnVault AI

Use these prompts before building. These help learners think clearly before opening Lovable. The goal is to avoid building a beautiful app that does not solve a real problem.

Prompt 1: Validate the Problem

Use this to confirm whether the learner problem is real.

I am validating a product idea.

Problem:

Learners attend courses, workshops, webinars, YouTube videos and classes. They save notes, links, screenshots and prompts in many places. Later, they forget what they learned, cannot find old notes easily and do not know what action to take next.

Help me validate whether this is a real problem.

Give me:

1. Who faces this problem
2. Why this problem matters
3. How often this problem may happen
4. What pain learners feel
5. Whether this problem is worth solving
6. A simple validated problem statement

Keep it simple and practical.

Prompt 2: Market Signals + Root Cause

Use this to find evidence and understand the deeper cause.

I want to understand whether people are already facing this problem and identify the real root cause.

Problem:

Learners save notes, links, screenshots, prompts and ideas in different places. Later, they struggle to find, revise, connect and apply what they learned.

Search for market signals from Reddit, forums, blogs, YouTube discussions and reviews of note-taking or learning apps.

Then use the 5 Whys method to find the deeper root cause.

Give me:

1. Key market signals
2. Common user complaints
3. Existing tools and workarounds
4. Gaps in current tools
5. Surface symptoms
6. 5 Whys analysis
7. Likely root cause
8. Product opportunity

Clearly separate symptoms from root cause.

Prompt 3: Target User + Product Solution

Use this to connect user needs to the right solution.

I want to define the target user and product solution for this idea.

Problem:

Learners save notes from courses, workshops, videos, articles and books, but later struggle to find, revise, connect and apply what they learned.

Root cause:

They do not have a simple system that turns captured learning into organized, useful knowledge and next actions.

Help me identify:

1. The primary target user
2. Their goals, current behaviour and main frustrations
3. The main product solution
4. The core product promise
5. The simplest user flow
6. Possible features
7. Which features solve the root cause
8. Which features should be included in Version 1

Focus on helping learners capture, understand, find, revise and apply learning.

End with a one-line product description.

Prompt 4: Prioritize Features with SAGE

Use this to choose useful V1 features.

We are building LearnVault AI, an AI second brain for learners.

Root cause:

Learners do not have a simple system that converts captured learning into organized memory, understanding and next action.

Possible features:

1. Learner dashboard
2. Add learning notes
3. Upload file or add link
4. Notes library
5. Search notes
6. Priority level
7. AI summary
8. Key takeaways
9. AI task generator
10. AI quiz from notes
11. Learning Buddy assistant
12. Weekly recap
13. Reminder to revise

Use SAGE to prioritize:

S = Solves root cause

A = Assists next action

G = Gives repeated value

E = Easy to build first

Score each feature out of 4 and recommend Version 1 features for a live workshop demo.

Prompt 5: Choose the WOW Feature

Use this to identify the memorable AI feature.

Help me identify the WOW feature for LearnVault AI.

A WOW feature should:

1. Feel impressive immediately

2. Create a useful result
3. Clearly show AI value
4. Solve an important learner problem
5. Give the user a reason to come back

Feature candidates:

- AI summary
- Key takeaways
- AI task generator
- AI quiz from notes
- Ask questions from notes
- Learning Buddy assistant

Classify each as essential utility, useful AI feature or WOW feature.

Recommend one main WOW feature and explain how to demonstrate it live.

Prompt 6: Mini Product Specification

Use this to create the Lovable build brief.

Create a clear mini product specification for LearnVault AI.

Product:

LearnVault AI

Positioning:

An AI second brain for learners.

Core promise:

Turn scattered learning notes into organized summaries, key takeaways, next actions, quizzes and an interactive learning companion.

Version 1 features:

1. Learner dashboard
2. Add notes by text, file upload or link
3. Save notes with title, topic, date and priority
4. Notes library
5. Search and filter notes
6. AI summary
7. Key takeaways
8. AI task generator
9. AI quiz from notes
10. Learning Buddy assistant

Give me:

1. Product overview
2. Target user
3. Main user flow
4. Screens required
5. Database requirements
6. AI features
7. API or integration needs
8. Version 1 exclusions
9. Clear instructions for Lovable

Keep it simple enough for non-technical learners.

5. Lovable Build Prompts

Use these prompts inside Lovable. Paste one prompt at a time, review the output, test it, and then continue. Do not paste all prompts at once.

Global UI Direction

Make LearnVault AI look modern, futuristic and premium: dark background, neon green accents, beautiful cards, strong typography, smooth spacing, AI loading states and mobile responsiveness.

Build Prompt 1: Foundation + Dashboard + Database

Build a premium web app called LearnVault AI using Lovable Cloud.

LearnVault AI is an AI second brain for learners. It helps learners save notes from workshops, courses, links and files, then turn them into summaries, key takeaways, next actions, quizzes and learning progress.

Design style:

Use a futuristic dark UI with neon green accents, premium cards, smooth spacing, strong visual hierarchy and beginner-friendly microcopy.

Create the Learner Progress Dashboard with:

1. Total notes saved
2. High-priority notes
3. This week's learning topics
4. Pending learning tasks
5. Quiz readiness score
6. Recent notes
7. Recommended next action
8. Button to add a new note
9. Button to ask Learning Buddy

Create database tables:

1. notes: id, title, content, source_type, file_url, link_url, topic, note_date, priority, ai_summary, key_takeaways, created_at, updated_at
2. learning_tasks: id, note_id, task_title, task_description, priority, status, created_at
3. quizzes: id, note_id, quiz_title, score, created_at
4. quiz_questions: id, quiz_id, question, options, correct_answer, explanation
5. buddy_conversations: id, question, answer, created_at

Make the app work even if some AI responses use sample generated text.

Build Prompt 2: Notes + Search

Add the notes feature and search feature.

Users should be able to add a learning note in three ways:

1. Paste text
2. Upload a file
3. Add a link

The Add Note form should include:

1. Workshop or session title
2. Topic
3. Date of note
4. Priority: high, medium, low
5. Note content
6. File upload option
7. Link input option

8. Save note button

After saving, the note should appear in the Notes Library.

Notes Library should include:

1. Search by workshop or session title
2. Search by note content or topic
3. Filter by date
4. Filter by priority
5. Cards showing title, topic, date, priority and short preview
6. Button to open note details

Keep the UI premium, futuristic and easy to use.

Build Prompt 3: AI Summary

Add an AI Summary feature for each saved note.

When the user opens a note, add a button called Generate AI Summary.

The AI summary should explain the note in a simple learner-friendly way and include extra context so the learner understands the topic quickly.

Output structure:

1. Simple summary
2. What this note is mainly about
3. Why this matters
4. Simple explanation for a beginner

Show the output in a beautiful AI Summary card.

Use structured output and friendly error handling.

Build Prompt 4: Key Takeaways

Add a Key Takeaways feature for each saved note.

When the user opens a note, show a button called Generate Key Takeaways.

The app should extract:

1. Five to seven key takeaways
2. Important concepts
3. Important terms to remember
4. One line on how the learner can apply this learning

Make the section clean, scannable and visually premium.

Build Prompt 5: AI Task Generator

Add an AI Task Generator feature.

This is the main WOW feature of LearnVault AI. The goal is to convert learning notes into clear next actions.

When the user opens a note, show a button called Generate My Next Tasks.

The app should generate:

1. Three practical next actions
2. Priority for each task: high, medium, low
3. Why this task matters
4. Suggested deadline
5. Button to mark task as completed

Also show a dashboard section called Pending Learning Tasks.

The task generator should feel like a learning coach telling the user what to do next.

Build Prompt 6: AI Quiz From Notes

Add an AI Quiz feature based on saved notes.

When the user opens a note, show a button called Quiz Me From This Note.

The quiz should include:

1. Five questions
2. Multiple choice answers
3. Correct answer after submission
4. Short explanation for each answer
5. Final score
6. Save quiz result to dashboard

Add a dashboard card called Quiz Readiness Score.

Make the quiz interactive, clean and fun without clutter.

Build Prompt 7: Learning Buddy Assistant

Add a Learning Buddy assistant feature.

Learning Buddy should help the learner ask questions about saved notes using text, and prepare the UI for future voice support.

Create a Learning Buddy screen with:

1. Text input
2. Voice input placeholder
3. Chat-style answer area
4. Suggested questions
5. Note references used in the answer

Suggested questions:

1. What did I learn this week?
2. What was important in the vibe coding session?
3. What is my next step?
4. What should I revise?
5. Quiz me from my notes

If full voice integration is not ready, clearly show it as a future integration placeholder.

Keep the assistant futuristic, friendly and impressive.

Build Prompt 8: Final Polish

Make the app demo-ready.

Improve:

1. Premium futuristic UI
2. Better spacing
3. Better cards
4. Smooth dashboard layout
5. Clear buttons
6. Better empty states
7. Clear note detail page
8. Better AI output sections
9. Mobile responsiveness
10. Consistent typography

Make the user journey clear:

Dashboard -> Add Note -> Search Notes -> Generate Summary -> Key Takeaways -> Generate Tasks -> Take Quiz -> Ask Learning Buddy.

Do not add unnecessary features.

Build Prompt 9: Test Complete Journey

Test the complete LearnVault AI user journey.

Check these flows:

1. User can add a note by pasting text
2. User can upload a file or add a link
3. User can save title, date and priority
4. User can search notes by title
5. User can filter notes by date and priority
6. User can generate AI summary
7. User can generate key takeaways
8. User can generate learning tasks
9. User can take a quiz from notes
10. User can open Learning Buddy and ask questions
11. Dashboard updates with notes, tasks and quiz score

Fix broken flows, missing buttons, confusing labels, poor UI spacing and AI structured-output errors.

Keep the design premium and simple.

6. Context Engineering Templates

Template A: Product Thinking Prompt

```
Act as a product strategist.

I am building [PRODUCT NAME].

Context:
[Who the user is]
[What problem they face]
[Why the problem matters]
[Current tools or workarounds]

Goal:
Help me define the right product solution.

Give me:
1. Target user
2. Pain points
3. Root cause
4. Product promise
5. Version 1 features
6. What to avoid

Keep it practical and beginner-friendly.
```

Template B: Lovable Build Prompt

```
Build [FEATURE NAME] for [APP NAME].

Context:
[Explain what the app does]
[Explain what this feature should help the user do]

Requirements:
1. [Requirement 1]
2. [Requirement 2]
3. [Requirement 3]

Database:
[Fields or tables needed]

UI:
[Design style and components]

Rules:
- Keep the UI clean and modern
- Add loading, empty and error states
- Do not break existing features
- Use real data where possible
- Make it mobile responsive
```

Template C: AI Structured Output Prompt

```
Use strict structured output.

Return only this object:
{
  "summary": "string",
  "key_takeaways": ["string"],
  "action_items": ["string"],
```

```

    "tags": ["string"],
    "revision_questions": ["string"]
  }

```

Rules:

1. Return only valid JSON/object output
2. Do not add markdown
3. Do not add explanation outside the object
4. Every field is required
5. Arrays must contain strings
6. If content is short, still generate useful values

7. Common Prompting Mistakes

Mistake	What happens	Better approach
Prompt is too vague	AI builds something random	Give user, problem, goal and feature list
Too many features at once	App becomes broken or confusing	Build one feature at a time
No output format	AI gives messy response	Ask for sections, tables or structured output
No constraints	AI adds unnecessary features	Clearly say what not to build
No testing prompt	Broken flows remain hidden	Ask Lovable to test the full journey

8. Final Learner Checklist

- I explained the user problem clearly.
- I gave AI enough context before asking for output.
- I separated symptoms from root cause.
- I prioritized features before building.
- I built one feature at a time in Lovable.
- I checked that data is saved correctly.
- I tested AI summary, key takeaways, tasks and quiz.
- I used structured output for AI features.
- I polished the UI for a strong demo.
- I tested the full user journey before presenting.